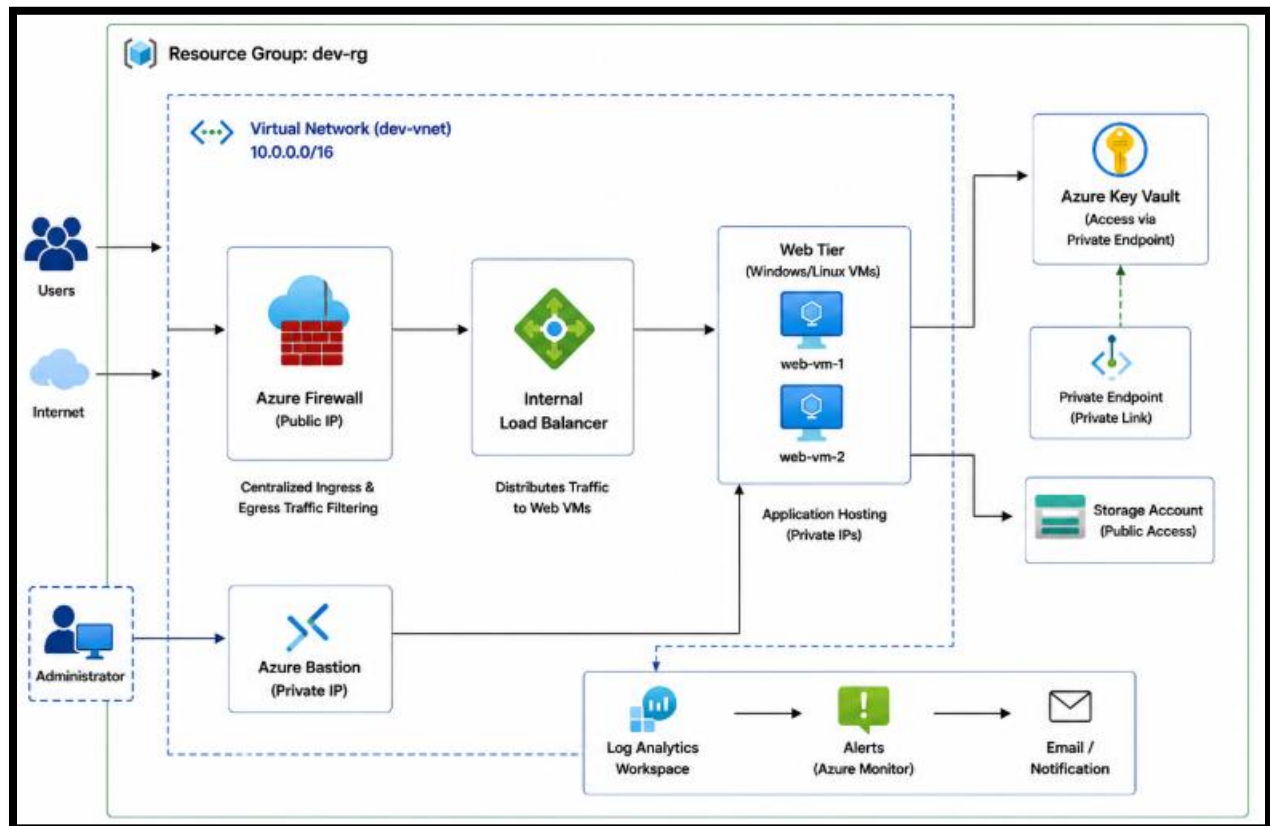


HIGH LEVEL DESIGN (HLD)

1. Overview

This architecture represents a secure and production-style Azure infrastructure built using Terraform modular architecture.



The environment is designed with:

- Secure inbound and outbound traffic control
- Private backend infrastructure
- Bastion-based administrative access
- Centralized monitoring and alerting
- Secure secret management using Key Vault

2. Architecture Components

Component	Purpose
Azure Firewall	Centralized traffic filtering
Internal Load Balancer	Backend traffic distribution
Web Tier (Linux VMs)	Application hosting
Azure Bastion	Secure VM administration
Azure Key Vault	Secret management
Private Endpoint	Secure Key Vault connectivity
Storage Account	Application and infrastructure storage
Log Analytics Workspace	Centralized logging
Azure Monitor Alerts	Monitoring and notifications

3. High Level Traffic Flow

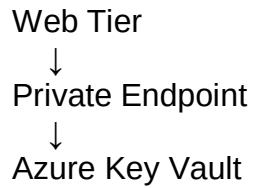
User Traffic

Users
↓
Internet
↓
Azure Firewall
↓
Internal Load Balancer
↓
Web Tier (Linux VMs)

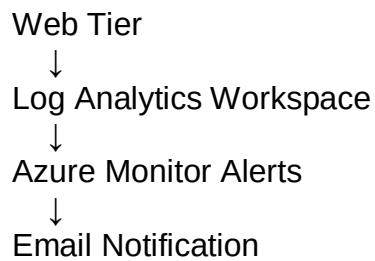
Administrative Access

Administrator
↓
Azure Bastion
↓
Web Tier (Linux VMs)

Secret Management



Monitoring Flow



4. Core Design Principles

- Private backend infrastructure
 - No public IP on virtual machines
 - Secure ingress and egress traffic
 - Bastion-based administrative access
 - Private connectivity to Key Vault
 - Centralized monitoring and alerting
 - Modular Terraform deployment model
-

5. Terraform Architecture

```
modules/  
├── compute/  
├── network/  
├── security/  
├── monitoring/  
└── storage-account/
```

6. Conclusion

This High Level Design provides a secure, scalable, and production-style Azure infrastructure using Terraform with centralized security, private networking, monitoring integration, and secure administrative access.